

Multi-Task Brain Tumor Segmentation and Classification

1. Introduction and motivation

Brain tumors are among the most serious neurological conditions, and accurate early diagnosis directly determines how a patient is treated. Magnetic Resonance Imaging (MRI) is the standard tool for diagnosis, but manual analysis of MRI scans is time-consuming, requires significant clinical expertise, and shows variability between readers. Automated methods that can detect, locate, and classify brain tumors therefore have real potential to assist clinicians and improve diagnostic consistency.

Most existing deep-learning work treats tumor localization (segmentation) and classification as two separate problems with two separate models. This is inefficient: both tasks examine the same image for the same abnormality, and the visual features useful for one are likely useful for the other. The theoretical case for solving them jointly was made by *Caruana (1997)*, and *Mlynarski et al. (2019)* demonstrated the principle directly on brain tumors. We follow this line of work and design a single unified model with a shared encoder feeding a segmentation decoder and a classification head, so that learning the two tasks together produces richer representations than learning either in isolation. The four categories of interest are glioma, meningioma, pituitary tumor, and no tumor.

A second challenge in medical imaging is that labeled data is expensive. To make better use of unlabeled images, we incorporate self-supervised pre-training: the encoder is first trained to reconstruct randomly masked image patches, following the Masked Autoencoder (MAE) framework of *He et al. (2022)*. The pre-trained encoder is then fine-tuned on the labeled multi-task objective.

2. Data description

All experiments use the BRISC2025 dataset (*Fateh et al., 2026*), a publicly available benchmark released for brain tumor analysis. The dataset contains around 6,000 expert-annotated MRI brain scans across four classes: glioma, meningioma, pituitary tumor, and no tumor. Each scan is a 2D grayscale slice drawn from a volumetric MRI sequence, and all annotations were produced by medical experts.

2.1 Dataset structure

The dataset is organized into two folders. The segmentation folder contains MRI images paired with binary pixel-level masks, where pixel value 1 indicates tumor tissue and 0 indicates healthy or background tissue. The classification folder contains the same MRI images labeled by tumor type, providing a single class label per image. Crucially, each image appears in both folders and the annotations can be aligned by filename.

For our unified model we combine both folders to construct training examples that carry both types of supervision: a segmentation mask and a class label per image. The "no tumor" class has no associated mask because there is no tumor to outline. We treat its mask as an all-zero array and the segmentation loss for these images is computed as the model successfully predicts the absence of tumor pixels.

2.2 Preprocessing

All images are resized to 224×224 pixels and normalized using **z-score normalization**: each pixel intensity is converted by $(\mathbf{x} - \boldsymbol{\mu}) / \boldsymbol{\sigma}$ using the mean and standard deviation computed over the training set only. This avoids any leakage from validation or test sets. Z-score normalization is preferred over fixed $[0, 1]$ scaling because MRI intensities have no fixed scale across scanners and patients, unlike natural photographs.

We apply standard data augmentation during training: random horizontal flips, small random rotations, and small contrast jitter. Augmentation is applied before any masking operation in the SSL stage. Lateral flipping is biologically valid because, while individual brains are not laterally symmetric, tumors are equally likely to occur on either hemisphere across the population.

The dataset comes with predefined train and test folders. We carve a 15% validation slice from the training data using a class-stratified split, ensuring all four classes are proportionally represented. The training set is used for both the SSL pre-training step and the supervised fine-tuning. The validation set is used for model selection, hyperparameter tuning, and early stopping. The test set is held out and only touched once at final evaluation, which protects against test-set tuning. We also performed structural sanity checks on the training data, counting corrupted images, missing tumor masks, and empty masks, before any model training, following the principle that failures discovered before training are much cheaper than failures discovered at epoch 30.

3. Method and architecture

We use a two-stage training methodology. Stage 1 is self-supervised pre-training of the encoder via masked image reconstruction. Stage 2 is supervised multi-task fine-tuning, where the pre-trained encoder is paired with a segmentation decoder and a classification head and trained jointly on labeled examples.

3.1 Stage 1 - Self-supervised pre-training (MAE)

We adapt the Masked Autoencoder framework of *He et al. (2022)*. Each 224×224 image is divided into a 14×14 grid of 16×16 patches. A random 60% of these patches are masked, and the encoder must produce features good enough that a small reconstruction decoder can predict the masked content. The reconstruction target is a per-patch normalized version of the original image, and the loss is the mean-squared error computed only on the masked patches. Visible patches do not contribute to the loss because the model already sees them.

We use a mask ratio of 60% rather than the 75% used in the original MAE paper. This deviation is deliberate. The 75% number assumes ImageNet-scale pre-training data and a Vision Transformer encoder that can sparsely process only the unmasked tokens. Both assumptions fail in our setup because we have approximately 4,250 training images (much smaller than ImageNet which has approximately 1.28M images) and a CNN encoder that processes the masked image densely. Subsequent literature on MAE for smaller and domain-specific datasets consistently uses lower mask ratios, and our preliminary runs supported this choice. The reconstruction decoder is intentionally lightweight and is discarded after pre-training. Only the encoder weights transfer to Stage 2.

3.2 Stage 2 - Multi-task fine-tuning

The fine-tuning model has three parts:

- **Encoder.** Loaded from Stage 1, fine-tuned during Stage 2.

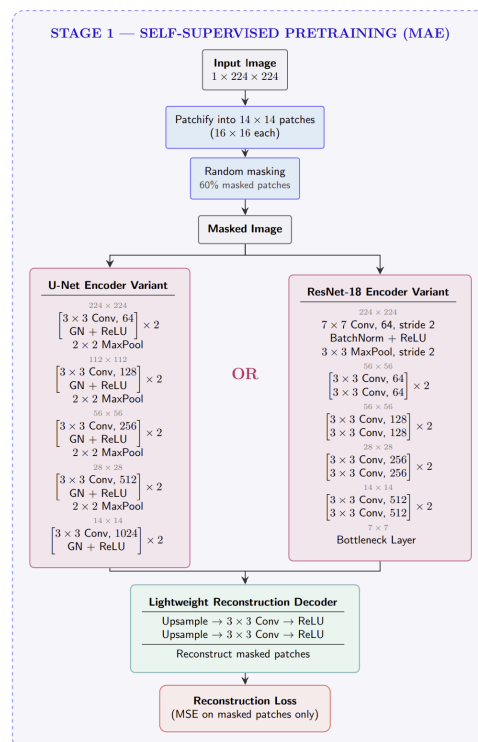
- **Segmentation decoder.** A U-Net-style decoder (*Ronneberger et al., 2015*) that takes the encoder bottleneck and the encoder skip connections and produces a $224 \times 224 \times 1$ logit map. The skip connections are essential: without them, the decoder would have to recover fine tumor boundaries from a heavily downsampled bottleneck, which is not possible.
- **Classification head.** A global average pooling layer followed by dropout and a linear layer that maps the bottleneck to four class logits.

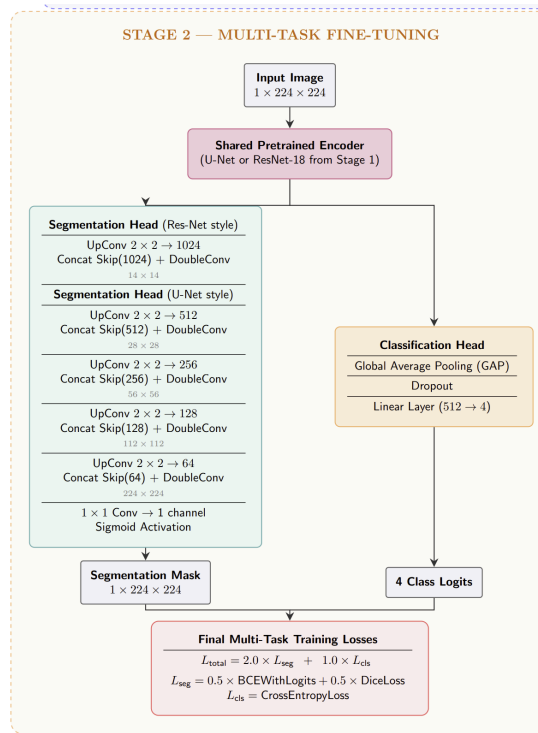
The two heads share the encoder. This is the multi-task hypothesis of Caruana (1997) made concrete: representations useful for "where is the tumor" overlap with representations useful for "what type of tumor", so forcing both heads to train through a shared encoder pushes it toward features that generalize across both tasks.

3.3 Encoder backbones

Following Chen (2023), we evaluate two encoder backbones under identical conditions.

- **U-Net encoder (baseline).** Five stages of a **DoubleConv** block (two 3×3 conv $\rightarrow$ **GroupNorm** $\rightarrow$ **ReLU** sequences) with **MaxPool2d** between stages. Channel counts $64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1024$. Spatial sizes $224 \rightarrow 112 \rightarrow 56 \rightarrow 28 \rightarrow 14$. The bottleneck is **(1024, 14, 14)**. Skip connections at four resolutions (224, 112, 56, 28) feed the decoder.
- **ResNet-18 encoder.** A standard **torchvision** ResNet-18 architecture (*He et al., 2016*) with the first convolution rebuilt for single-channel input. Spatial sizes go $224 \rightarrow 112 \rightarrow 56 \rightarrow 28 \rightarrow 14 \rightarrow 7$, which is one extra downsampling step compared to the U-Net encoder. Channel counts $64 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. The bottleneck is **(512, 7, 7)** (half the channels and a quarter of the spatial area of the U-Net bottleneck). Skip connections at four resolutions (112, 56, 28, 14) feed a matching decoder. The ResNet-18 decoder needs five upsampling stages to return to 224×224 (one more than the U-Net decoder), and its final stage has no skip connection because the encoder never produces a feature map at full input resolution.





3.4 Normalization choices

The U-Net encoder and both decoders use **GroupNorm** rather than BatchNorm. GroupNorm computes statistics within each individual sample, splitting channels into groups, so it is batch-size agnostic. BatchNorm becomes unstable at the small batch sizes we use, and the nnU-Net, uses GroupNorm by default for the same reason (Wu & He, 2018, discuss this trade-off in detail). The ResNet-18 encoder retains its native BatchNorm because replacing it would have been a confounded change that mixes "encoder choice" with "normalization choice"; we keep the standard backbone so the comparison is faithful to what Chen (2023) proposed.

3.5 Loss functions

For segmentation, we combine binary cross-entropy with logits and Dice loss:

$$\mathcal{L}_{\text{seg}} = \alpha \cdot \text{BCE}(p, y) + (1 - \alpha) \cdot \text{Dice}(p, y), \quad \alpha = 0.5$$

BCE provides stable per-pixel gradients but is sensitive to the class imbalance inherent in segmentation (most pixels are background). Dice is directly proportional to overlap and therefore robust to imbalance, but its gradients are unstable when predictions are near zero. Combining them lets each cover the other's weakness. We use **binary_cross_entropy_with_logits** (rather than sigmoid then BCE) for numerical stability.

For classification we use plain cross-entropy over the four classes. The total multi-task loss combines them with **fixed weights**:

$$\mathcal{L}_{\text{total}} = w_{\text{seg}} \cdot \mathcal{L}_{\text{seg}} + w_{\text{cls}} \cdot \mathcal{L}_{\text{cls}}, \quad w_{\text{seg}} = 2.0, w_{\text{cls}} = 1.0$$

We initially attempted *Kendall et al.*'s learned uncertainty weighting but found that the learned weights drifted in directions that gamed the total loss without improving the predictions themselves. We therefore reverted to fixed weights, with the segmentation loss weighted twice as heavily because it is the harder task. Segmentation requires per-pixel precision while classification only requires a single decision per image.

3.6 Evaluation metrics

We report four metrics:

- **Dice coefficient** (segmentation overlap), the clinical standard.
- **IoU** (Intersection over Union), monotonically related to Dice but more strict.
- **Hausdorff Distance** (boundary error), which captures the worst-case boundary mismatch in a way Dice cannot.
- **Macro F1** for classification, which weights all four classes equally regardless of frequency.

For checkpointing during fine-tuning we use a **composite metric**, $0.5 \cdot \text{Dice} + 0.5 \cdot \text{macro F1}$. . Selecting the best model on a single metric is fragile because a single epoch with high Dice but failed classification (or vice versa) could be picked as best. This approach of the composite metric ensures the saved checkpoint is balanced.

4. Results

In this section, we are not planning to put any figures about the results. All evaluation figures are included in the notebook.

4.1 Test-set comparison: U-Net vs ResNet-18

The two architectures are identical apart from the encoder. Both go through identical SSL pre-training and identical multi-task fine-tuning on identical data with identical seed (42). Any difference in test-set metrics is therefore attributable to the encoder change alone.

Metric	U-Net	ResNet18-Unet	Δ (UNet-ResNet18)
Composite	0.8827	0.8990	- 0.0163(ResNet)
Dice(avg)	0.8473	0.8383	+ 0.009(Unet)
IoU(avg)	0.7729	0.7615	+ 0.0114(Unet)
Hausdorff(avg, px)	10.33	11.70	- 1.47(Resnet)
Macro F1	0.9181	0.9597	- 0.0416(Resnet)
Accuracy	0.916	0.958	- 0.042(Unet)

Per-class Dice on the test set:

Class	U-Net Dice	ResNet-18
Glioma	0.7151	0.7018
Meningioma	0.9242	0.9273

Pituitary	0.8808	0.8632
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Parameter counts and inference cost:

Model	Total Parameters	Ms/Forward(single image)
U-Net	~31.0M	33.8
ResNet-18-Unet	~14.3M	5.6

The ResNet-18 model has approximately 46% of the U-Net's parameters, supporting the "lower-resource occupation" claim of Chen (2023).

4.2 SSL removed

To quantify the contribution of the SSL pre-training stage we ran an additional configuration where Stage 1 was skipped entirely (encoder initialized randomly, then taken directly into the multi-task fine-tuning stage). Results:

Configuration	Dice	Macro F1
U-Net+SSL	0.8473	0.9181
U-Net (no SSL)	0.8490	0.9603
ResNet-18+SSL	0.8383	0.9597
ResNet-18(no SSL)	0.8264	0.9670

5. Analysis and discussion

5.1 Did the encoder swap help?

The results show a split verdict. The U-Net encoder is slightly better at segmentation (Dice 0.847 vs 0.838, IoU 0.773 vs 0.762, Hausdorff 10.3 vs 11.7 pixels), while the ResNet-18 encoder is clearly better at classification (macro F1 0.960 vs 0.918, accuracy 95.8% vs 91.6%). The composite metric favours ResNet-18 (0.899 vs 0.883) because the classification gap is larger than the segmentation gap. This does not reproduce the clear segmentation advantage reported by Chen (2023), who found a 5-point IoU improvement when switching from U-Net to ResNet-18. The most likely explanation is that our setting differs from theirs in ways that reduce the encoder's relative importance. First, our SSL pre-training gives the U-Net encoder a strong initialization that partly closes the gap with ResNet's residual connections. Second, we train on a four-class multi-task problem rather than binary segmentation, which changes the gradient landscape the encoder sees. Third, the BatchNorm layers retained in the ResNet-18 backbone may be slightly unstable at our batch size of 16, offsetting some of the benefit of residual connections.

From a practical standpoint, the choice between the two encoders depends on what matters most in deployment. If segmentation accuracy is the primary goal, the U-Net encoder is the safer pick. If classification matters more, or if inference cost is a constraint, ResNet-18 is preferable, it runs 6x faster (5.6 ms vs 33.8 ms per image) with less than half the parameters (14.3M vs 31.0M).

5.2 Did SSL pre-training help?

The SSL ablation produced a surprising result: self-supervised pre-training did not clearly help on this dataset. For the U-Net encoder, removing SSL actually improved both Dice (0.849 vs 0.847) and macro F1 (0.960 vs 0.918). For ResNet-18, SSL gave a small Dice improvement (0.838 vs 0.826) but a small F1 decrease (0.960 vs 0.967).

The most likely explanation is that the BRISC2025 training set, with roughly 4,250 labeled images, is large enough for both encoders to learn good features directly from supervised training. The 100 epochs of SSL pre-training add significant compute but do not provide representations that the supervised stage cannot discover on its own. This is consistent with the broader MAE literature, which shows the largest SSL benefits either on very large unlabeled datasets or when the labeled fraction is very small, our setting has neither extreme.

This is itself a useful finding. SSL pre-training is not universally beneficial, and practitioners working with moderately sized labeled datasets should verify its contribution through an ablation rather than assuming it helps.

5.3 Per-class behavior

Glioma is the hardest class for both architectures, with Dice scores of 0.715 (U-Net) and 0.702 (ResNet-18). These are substantially lower than meningioma (0.924 / 0.927) and pituitary (0.881 / 0.863). This pattern is expected: gliomas tend to have irregular, diffuse boundaries that are difficult to delineate even for human readers, while meningiomas and pituitary tumors are typically more compact and well-defined.

The two encoders perform similarly across all three tumor classes, with differences of 1-2 percentage points and no consistent winner. This suggests that the slight overall segmentation advantage of U-Net over ResNet-18 is spread evenly across tumor types rather than driven by one class where U-Net is specifically better.

6. Conclusion

We presented a unified deep learning system for brain tumor segmentation and classification on the BRISC2025 dataset. The system combines self-supervised MAE pre-training with multi-task fine-tuning using a shared encoder, a U-Net-style segmentation decoder, and a classification head. We evaluated two encoder backbones, a standard U-Net contracting path and a ResNet-18 backbone following Chen (2023), under identical training conditions to isolate the effect of the encoder choice.

The comparison revealed a task-dependent trade-off: the U-Net encoder produced slightly better segmentation (Dice 0.847 vs 0.838), while the ResNet-18 encoder produced substantially better classification (macro F1 0.960 vs 0.918) at less than half the parameter count and 6x faster inference. The SSL ablation showed that self-supervised pre-training did not meaningfully improve results on this dataset, suggesting that the labeled training set is large enough for both architectures to learn strong features from supervised training alone.

The work demonstrates that multi-task learning is a practical strategy for joint brain tumor segmentation and classification, and that the encoder backbone is a meaningful design choice whose effect depends on which downstream task matters most.

References

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